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Research

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Anopheline Mosquito Species Composition, Kdr Mutation Frequency, and Parasite Infectivity Status in Northern Tanzania

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Abstract

The scaling-up of malaria control interventions in northern Tanzania has resulted in a decline in malaria prevalence and vector species composition. Despite this achievement, residual malaria transmission remains a concern in the area. The main aim of this study was to investigate malaria vector species composition, parasite infectivity rates, and the presence of insecticide knockdown resistance (kdr) mutations in three sites that have experienced a significant decline in malaria in northern Tanzania. Adult mosquitoes were sampled using light traps in houses and hand-aspirators in cowsheds, whereas the standard dipping method was used for sampling mosquito larvae. Adult mosquitoes identified as *Anopheles gambiae* s.l. and *An. funestus* s.l. and larval stages III and IV of *An. gambiae* s.l. were stored in absolute ethanol for further laboratory molecular identification. The identified species in the *An. gambiae* complex were *An. gambiae* s.s., *An. merus*, *An. quadriannulatus*, and *An. arabiensis*, whereas the *An. funestus* group comprised *An. funestus* s.s., *An. rivulorum*, and *An. leesoni*. For *An. gambiae* s.s. analyzed from Zeneth, 47.6% were kdr-East homozygous susceptible, 35.7% kdr-East heterozygous resistant, 9.6% kdr-East homozygous resistant, and 7.1% undefined, whereas specimens from Kwakibuyu were 45.5% kdr-East homozygous susceptible, 32.7% kdr-East heterozygous resistant, 16.3% kdr-East homozygous resistant, and 5.5% undefined. There were no kdr-West alleles identified from any specimen. The overall malaria parasite infectivity rate was 0.75%. No infections were found in Moshi. The findings indicate that populations of the major malaria vector mosquitoes are still present in the study area, with *An. funestus* taking a lead in malaria transmission.

Key words: vector, malaria, habitat, trapping, mutation

Understanding the distribution and species composition of malaria vector mosquitoes, knockdown resistance (kdr) mutation type, and *Plasmodium* parasite infectivity rates in Tanzania is of paramount importance in designing effective vector control tools and strategies. The vector populations decline across sub-Saharan Africa including Tanzania need special attention for the management of residual malaria transmission (Sokhna et al. 2013, Chinula et al. 2018, Ogwang et al. 2018). The confirmed vector species among the *Anopheles gambiae* complex existing in Tanzania are *An. gambiae* s.s., *An. merus* (in coastal areas), and *An. arabiensis* in savannah and semiarid areas of mainland (Matola and Magayuka 1981; Mnzava and Kilama 1986; Temu et al. 1998; Mboera and Magesa 2001; Kweka et al. 2008a,c; Mwanziva et al. 2011; Charlwood et al. 2018). Ecologically, *An. gambiae* and *An. funestus* were predominant within the areas characterized by high humidity and fresh water breeding bodies. *Anopheles merus* occupying the costal belt (Matola and Magayuka 1981; Mnzava and Kilama 1986; Temu et al. 1998; Mboera and Magesa 2001; Kweka et al. 2008a,c).

In recent years, there have been shifting in vector species proportions, *An. gambiae* s.s. population has been turned over with

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An. arabiensis increased population whereas *An. funestus* population has been considerably leading in malaria transmission in different areas of Tanzania, due to implementation of indoor residual house spraying and high coverage of long lasting insecticidal bed nets (LLINs; Russell et al. 2010, Derua et al. 2012, Kitau et al. 2012, Kulkarni et al. 2016), with pyrethroids as a main class of insecticide being used. The LLIN coverage and ownership in Tanzania has increased enormously since the initial scaling up in 2005 (Magesa et al. 2005), from 62.5 to 90.8% (Widmar et al. 2009, West et al. 2012, Stuck et al. 2017). The dependence on insecticide-based malaria vector control across many parts of Africa has been associated with the evolution of insecticide resistance in all the major vector species (Katureebe et al. 2016, Abeku et al. 2017, Echaubard et al. 2018).

The aim of this study was to assess mosquito species composition, kdr mutation frequencies, and parasite infectivity rates of malaria vectors in Northern Tanzania.

Materials and Methods

Study Sites

The study was conducted in three sites, Muheza, Moshi, and Magugu. Muheza comprises three villages: Genge (S 5.17170; E 38.79378, 139 m above sea level), Zeneth (S 5.22255; E 38.65767, 204 m above sea level), and Kwakibuyu (S 5.2225.18057; E 38.78828, 247 m above sea level) and is a malaria holoendemic area (Fig. 1). The main economic activity is small-scale agricultural farming of maize, cassava, rice, and beans. Moshi site is located in the foothills of Mt. Kilimanjaro, in Mabogini ward (S03°.21′, E37°.20′, 750 m above sea level; Fig. 1). The main income generating activities in this area include rice farming, maize, vegetable, and banana plantations. This study site has very low seasonal malaria transmission.

The Magugu site is located in the rift valley of northern Tanzania, south of Lake Manyara. The main economic activities are agriculture, which includes rice farming, maize, vegetable, and sugarcane plantations. This site has low malaria transmission and occasionally malaria epidemics (Mboera and Kitua 2001). Larvae were sampled from two hot springs: Mahole (S03.87437°, E 035.71528°, 972 m) and Masware (S04.06239°, E035.71503, 1,053 m; Fig. 1).

Mosquito Sampling Technique

Sampling was done twice a year, first on May after long rain season and second sampling was on September during dry season of 2015. Adult mosquitoes were collected using CDC miniature light traps as reported in previous studies (Sadanandane et al. 2004, Sriwichai et al. 2015). Adult mosquitoes were sampled in 10 houses per site for 4 wk consecutively after rain season and in dry season. Traps had no bait. Adults trapped were preserved on silica gel separated by cotton for further processing (WHO 1975). Species identification was done using a morphological key. All *An. gambiae* s.l. were stored for molecular analysis to species level (Coetzee et al. 2013) and sporozoite frequency (Wirtz et al. 1985). Mosquito larvae were sampled using the standard dipper (350 ml) in all habitats. The number of dipping varies with the size of habitat. First and second instar anopheline larvae were returned to their respective habitats, whereas third and fourth larval instars were taken to the laboratory for morphological identification and preservation in absolute ethanol for further analyses.

Anopheles gambiae s.l. Species Identification

All specimens of *An. gambiae* s.l were identified to species level by standard DNA PCR methods (Scott et al. 1993). DNA samples were

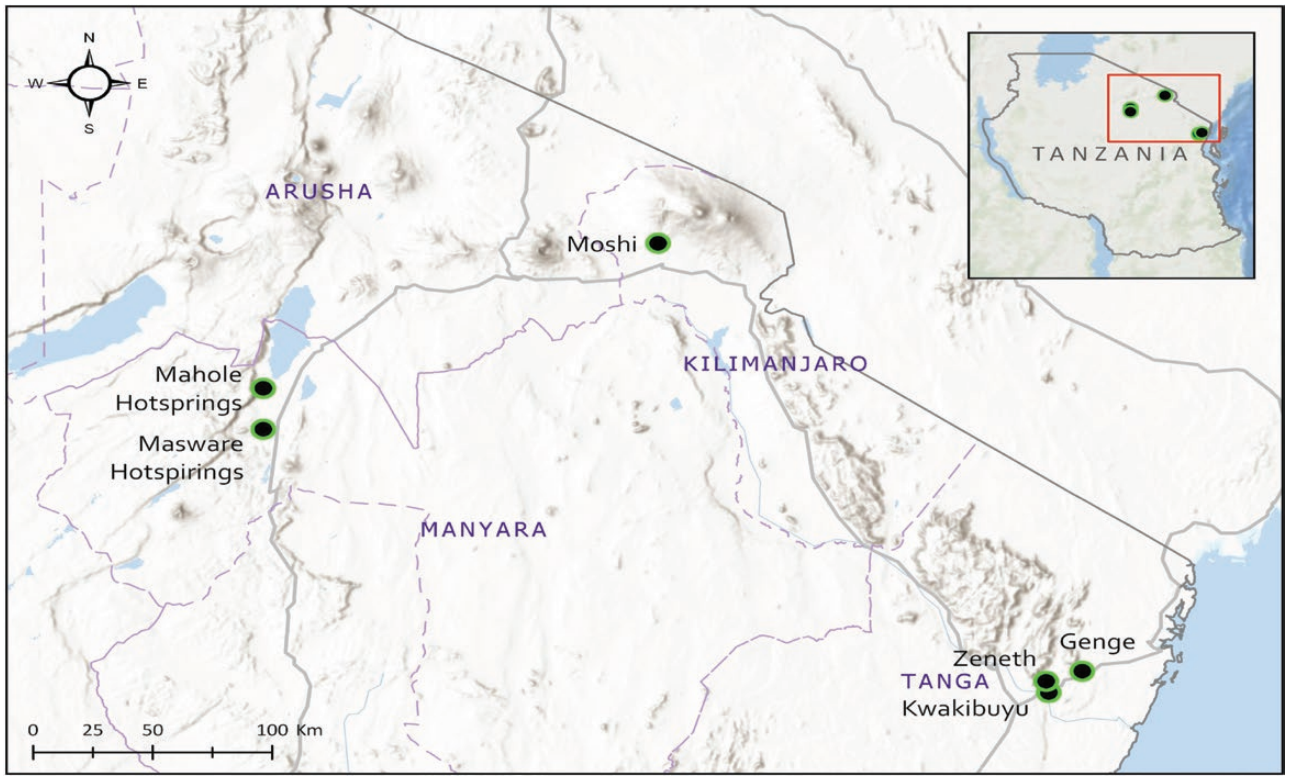


Fig. 1. Map showing the study sites where sampling of mosquitoes were done: Masware (larvae), Mahore (larvae), Moshi (adults), Genge (larvae), Zeneth (adults), and Kwakibuyu (adults).

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extracted from a single leg or wing or whole larva using a standard DNA extraction protocol (Collins et al. 1987).

Anopheles funestus Species Identification

Genomic DNA was extracted from a single leg or a wing using an extraction KIT (ZyGEM Corp Ltd., New Zealand) according to the manufacturer's instructions. The PCR methodology of (Koekemoer et al. 2002) was used.

Sporozoite ELISA

Head and thorax of adult female mosquitoes were individually homogenized in 50-µl grinding buffer (BB-NP-40) and pestle washed with 150 µl of blocking buffer to give a total volume of 200 µl per specimen. The methodology developed by Wirtz et al. (1985) was used for the sporozoite analysis.

Detection of kdr Mutation Types

The kdr mutation is a mechanism associated with resistance of pyrethroid-based insecticides in most of vector species, which is caused by point mutations in the pyrethroid target site called paratype sodium channel (Yewhalaw and Kweka 2016). The presence of kdr mutation in *An. gambiae* population was detected in allele L1014S and termed East African mutation (Kdr-E), whereas mutation in allele L1014F was termed as west Africa mutation (Kdr-W; Verhaeghen et al. 2006). The detection of East (kdr-E) and West (kdr-W) kdr mutation genotypes in *An. gambiae* s.s. from Northern Tanzania was performed using a previously reported PCR protocol (Ranson et al. 2000).

Results

Mosquito Species Composition by Sites

In total, 2,313 samples of *An. gambiae* s.l. and 422 *An. funestus* s.l. specimens were collected. Of these, 1,025 samples of *An. gambiae* s.l. and 324 *An. funestus* group were analyzed to species level (Table 1). *Anopheles gambiae* s.s., *An. arabiensis*, *An. funestus* s.s., *An. rivulorum*, and *An. leesoni* were identified from Muheza; only *An. arabiensis* identified from Moshi; and *An. arabiensis*, *An. quadriannulatus*, and *An. merus* from the Magugu hot springs (Table 1). Eighty samples from the Magugu hot springs were not amplified by using the Scott et al.'s (1993) PCR assay for *An. gambiae* complex.

Infectivity Rates

In total, 300 *An. gambiae* s.l. and 100 *An. funestus* s.l. were analyzed with ELISA for *Plasmodium falciparum* circumsporozoite protein. Out of these, one *An. gambiae* s.s. from Kwakibuyu and two *An. funestus* s.s. (Kwakibuyu and Zeneth each) were found to be positive for circumsporozoite protein (Table 1). Hence, the overall infectivity rate was 0.75% for Muheza. No infected *An. arabiensis* were found in Moshi.

Kdr (East and West) Mutations Detection

In total, 97 specimens of *An. gambiae* s.s. were subjected to kdr East/West analyses of which 42 specimens of *An. gambiae* s.s. were from Zeneth and 55 were from Kwakibuyu. For specimens from Zeneth, 47.6% ($n = 20$) were kdr-East homozygous susceptible, 35.7% ($n = 15$) kdr-East heterozygous resistant, 9.6% ($n = 4$) kdr-East homozygous resistant, and 7.1% ($n = 3$) were undefined, whereas for specimens from Kwakibuyu, 45.5% ($n = 25$) were kdr-East homozygous susceptible, 32.7% ($n = 18$) kdr-East heterozygous resistant, 16.3% ($n = 9$)

Table 1. The species identification and infectivity rates of mosquitoes samples from three sites Muheza (Kwakibuyu, Genge, and Zeneth), Moshi, and Hotspring (Masware and Mahore)

			Number of mosquitoes tested	*Anopheles gambiae* s.s.	*Anopheles arabiensis*	*Anopheles quadriannulatus*	*Anopheles merus*	*Anopheles rivulorum*	*Anopheles leesoni*	*Anopheles funestus* s.s.	Samples without amplification
Species	Kwakibuyu	*An. gambiae* s.l.	159	122	37	—	—	—	—	—	—
		An. funestus	163	—	—	—	—	18	25	120	—
	Zeneth	*An. gambiae* s.l.	150	110	40	—	—	—	—	—	—
		An. funestus	161	—	—	—	—	16	7	138	—
	Genge	*An. gambiae* s.l.	100	10	90	—	—	—	—	—	—
	Moshi	*An. gambiae* s.l.	300	—	300	—	—	—	—	—	—
	Hot spring (Masware & Mahore)	*An. gambiae* s.l.	316	—	191	32	13	—	—	—	80
Infectivity	Kwakibuyu	*An. gambiae* s.l.	100	1	—	—	—	—	—	—	—
		An. funestus	50	—	—	—	—	—	—	1	—
	Zeneth	*An. gambiae* s.l.	100	—	—	—	—	—	—	—	—
		An. funestus	50	—	—	—	—	—	—	1	—
	Moshi	*An. gambiae* s.l.	100	—	—	—	—	—	—	—	—

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kdr-East homozygous resistant, and 5.5% (n = 8) were undefined (Fig. 2). The frequency of the kdr-East mutation was 0.3 at both localities. Tests for Hardy–Weinberg equilibrium gave a χ^2 value of 0.22, which is insignificant ($P > 0.05$). There was no identification of the kdr-West mutation from any of the specimens analyzed.

Discussion

The findings of this study have shown that regular surveys are important to update the vector species presence and profile. It has confirmed the presence of *An. quadriannulatus* for the first time in northern Tanzania from the hot springs in Magugu, south of Lake Manyara. Previously, this species had only been recorded from Zanzibar Island and Mbozi in south-western Tanzania. *Anopheles merus* was collected from the same hot springs, which is the first report of this species so far in mainland, previously known from Zanzibar islands and various localities on the mainland. The 80 larvae that were not identified by the *An. gambiae* complex PCR assay were also not identified morphologically as belonging to this complex, so presumably these were other anopheline species and this requires molecular confirmation. Unlike the hot springs of Bwamba forest in Uganda with similar characteristics, where the main vector found was *An. bwambae* (Townson and Onapa 1994), no larvae of this species were found. *Anopheles quadriannulatus* is mainly zoophilic whereas *An. merus* is anthropogenic and its presence might explain the high malaria transmission in Magugu area, leading to recorded epidemics (Mboera and Kitua 2001).

Muheza showed the greatest species diversity with five species being identified, two members of the *An. gambiae* complex and three members of the *An. funestus* group. The major malaria vector species were the most abundant with 258 *An. funestus* s.s., 242 *An. gambiae* s.s., and 167 *An. arabiensis*. This is similar to previous studies carried out at the same locality (Mboera and Magesa 2001, Kweka et al. 2008a).

All specimens from Moshi were identified as *An. arabiensis*, which is similar to previous studies (Mahande et al. 2012b, Mbepera et al. 2017, Nnko et al. 2017). This species abundance in the area has been consistently associated with low seasonal malaria transmission (Oesterholt et al. 2006, Shekalaghe et al. 2007, Shekalaghe et al. 2011).

In this study, *An. funestus* group species were found only in Muheza, at Zeneth and Kwakibuyu. In both sites, *An. funestus* s.s. was the most prevalent species, with *An. leesoni* and *An. rivulorum* occurring in much lower numbers. Similar results were found in the coastal region of Bagamoyo, where *An. funestus* s.s. and *An. leesoni* were found feeding on human blood and were positive for circumsporozoite protein, whereas *An. rivulorum* was found not to have taken a human bloodmeal (Temu et al. 2007). Recently, in South Africa, an observation indicated that *An. vaneedeni*, which has never been found to be parasite positive in nature, was confirmed positive for *Plasmodium falciparum*, indicating the expanded role of members of the *An. funestus* species group in malaria transmission (Burke et al. 2017). Similar studies conducted at sites in the Muheza area also identified *An. funestus* s.s. as the active malaria vector (Mboera and Kitua 2001, Kweka et al. 2008b, Derua et al. 2015).

The overall parasite infectivity rate of the collected malaria vectors was 0.75% with infections found in *An. gambiae* s.s. (1/300) and *An. funestus* s.s. (2/100). Compared with previous findings, the infectivity rates of the same malaria vectors have declined significantly in the same area (Mboera and Magesa 2001, Kweka et al. 2008a, Kabula et al. 2011, Derua et al. 2012). The decline of malaria transmission in this study site has been attributed to several factors, such as scaling up of long lasting insecticidal nets, increased coverage of Indoor residual spray and community sensitization on use of malaria vector control measures such as house modifications and other technological tools (Mmbando et al. 2010, Ishengoma et al. 2011, West et al. 2015, Francis et al. 2017, von Seidlein et al. 2017).

The current findings indicate that the East African insecticide knockdown resistance (kdr-East) mutation is prevalent in *An. gambiae* s.s. in the Muheza area. The frequency of kdr-East was 0.3 with no deviation from the expected Hardy–Weinberg equilibrium ($P > 0.05$). There were no kdr-West alleles found in these *An.*

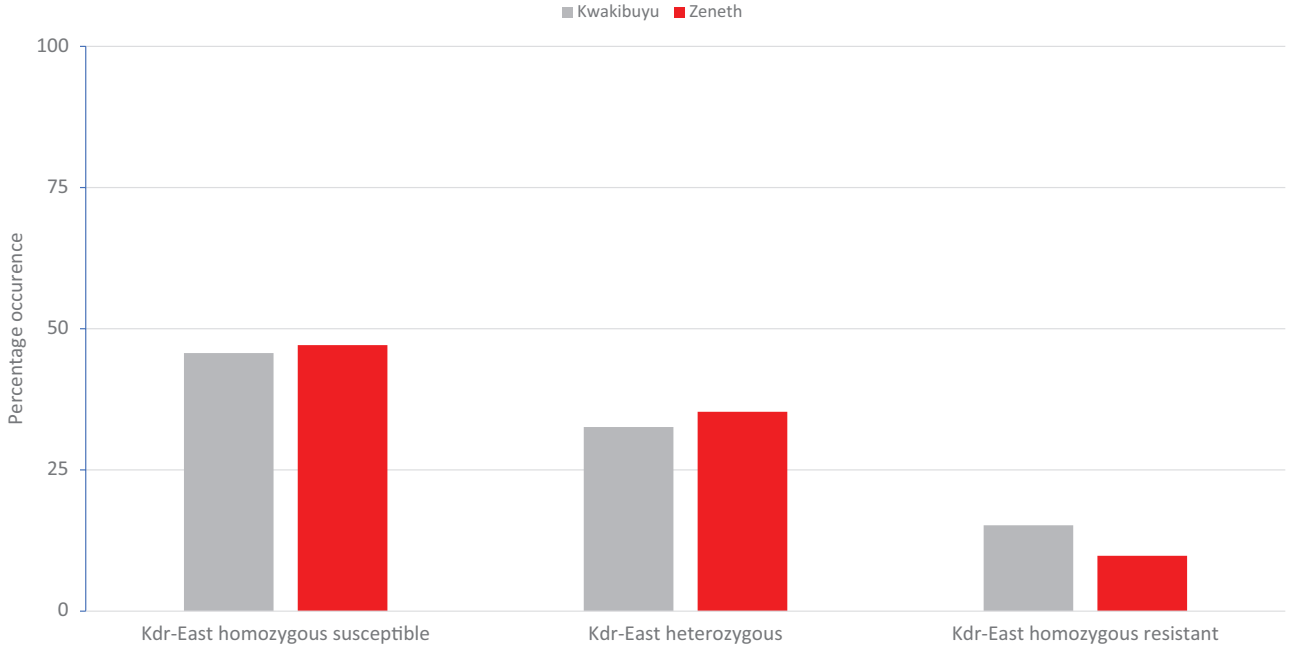


Fig. 2. The frequency of knockdown resistance (kdr) mutations of East Africa type.

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gambiae s.s. populations. A previous study carried out in Muheza in 2011 found no kdr mutations in five *An. gambiae* s.s. This apparent increase in frequency of the kdr-East mutation has also been noted in other malaria endemic areas (Kulkarni et al. 2006, Mahande et al. 2012a, Kabula et al. 2016, Kisinza et al. 2017, Mbepera et al. 2017, Nnko et al. 2017, Wanjala and Kweka 2018).

This study conducted in three different ecological settings highlights the importance of entomological surveillance for vector control, particularly in areas where malaria transmission and vector populations have declined substantially. These areas should be monitored in terms of malaria vector species composition, insecticide resistance profiles, the insecticide resistance mechanisms, and malaria transmission rates, and, last, to understand the feeding and resting behavior of the vectors.

Conclusion

The findings indicate that there is an increase of kdr mutations in the malaria vector *An. gambiae* s.s., in the areas which have been well covered by malaria vector control interventions and reporting significant declines in malaria prevalence.

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